

Salus Documentation

In today's fast-paced world, managing healthcare data efficiently is a persistent challenge. Patients struggle with fragmented medical records, hindering their ability to track progress or understand complex reports, while doctors lack unified access to patient histories, slowing diagnostics and treatment.

Meet Salus—named after the Roman goddess of well-being and safety—a revolutionary AI-powered platform that solves these issues. Salus securely centralizes health data, simplifies reports into clear insights, and facilitates seamless doctor-patient collaboration, all while prioritizing privacy.

Harnessing advanced data processing and generative AI, Salus delivers a scalable, intelligent solution—offering medication tracking, appointment scheduling, AI-driven diagnostics, and comprehensive health analysis to empower better outcomes for all.

We aim to connect with hospitals and launch our application with their doctors and patients connected once we are done , we will move to different hospital but in the same geo-location.

- **Tech Used**

- Frontend: React JS
- Database: SQL (hosted on snowflake)
- AI model: Python, Snowflake

Salus has two different interfaces for doctor and patients each providing different set of functionalities to both of them

- **Features offered to patients:**

- Medication alert
- Long term health tracker
- Detailed insight of health report
- Booking & tracking appointments
- Sharing health record with doctor

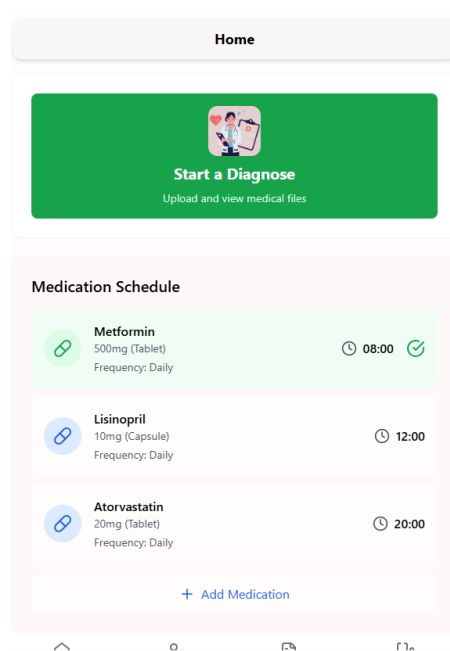


Fig.1 Home page with medication alert

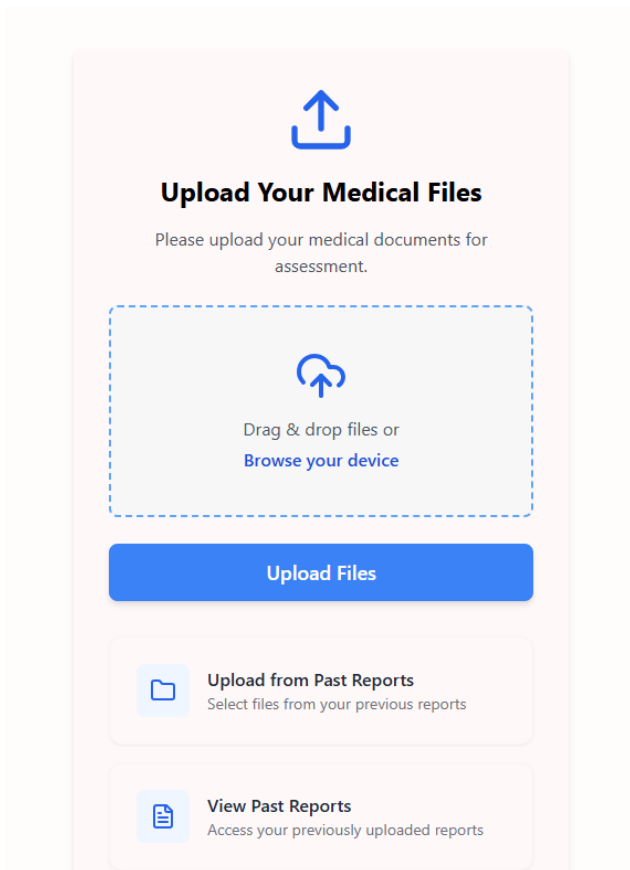


Fig.2 Upload your health report

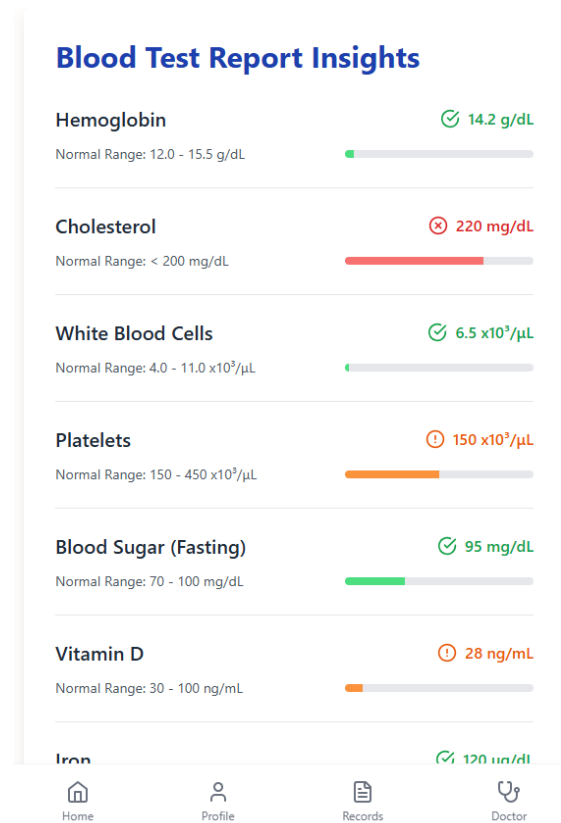


Fig3. Simplified insight of blood report

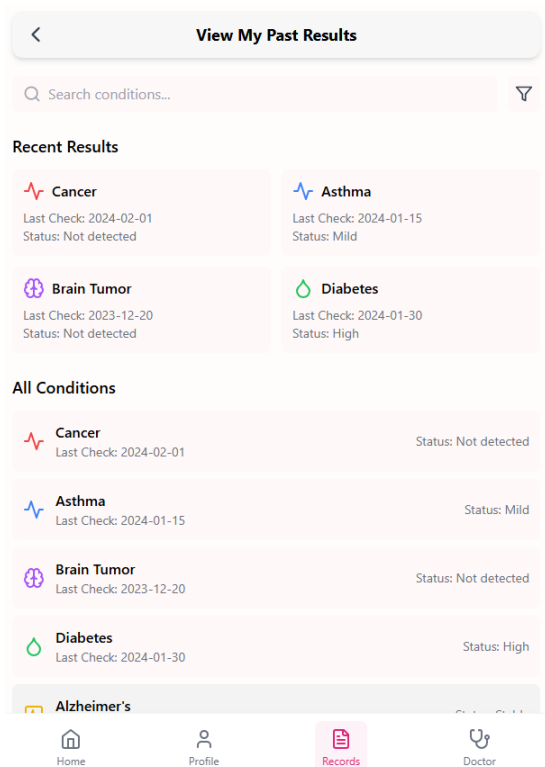


Fig4. Page to see health metrics

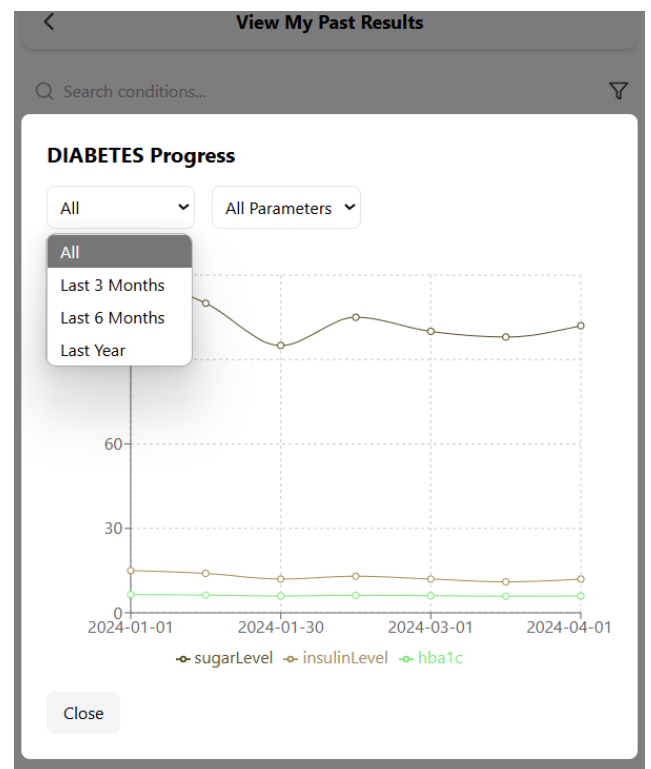


Fig5. Visual representation of parameters Over a period of time

DIABETES Progress

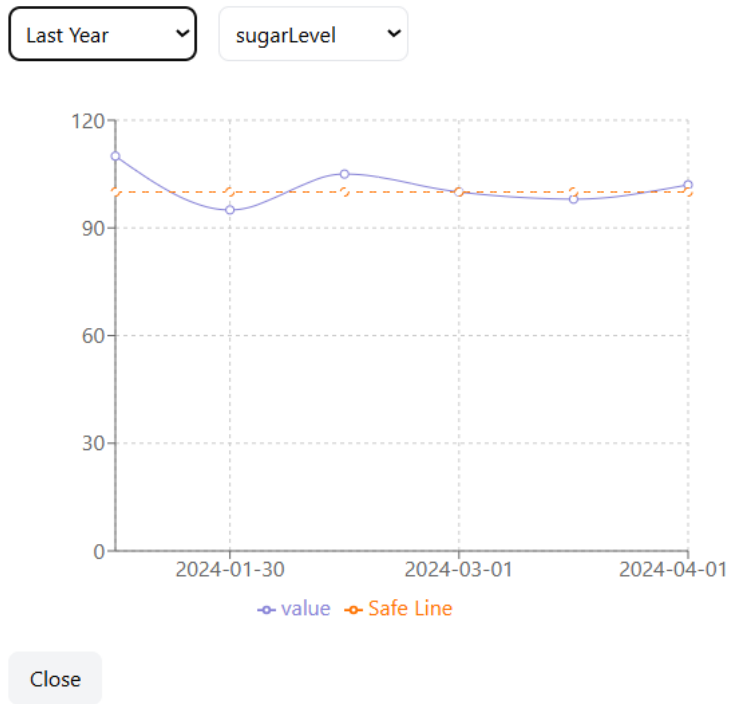


Fig6. Parameter-wise visualization

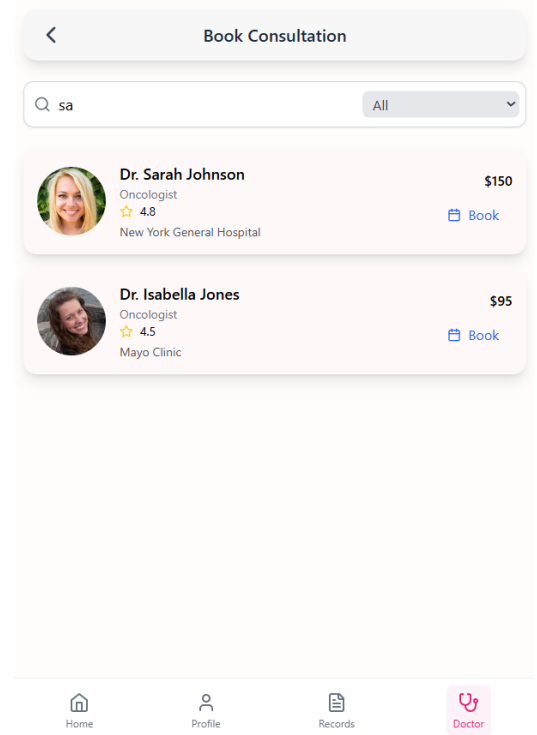


Fig7. Page to book appointments with doctors

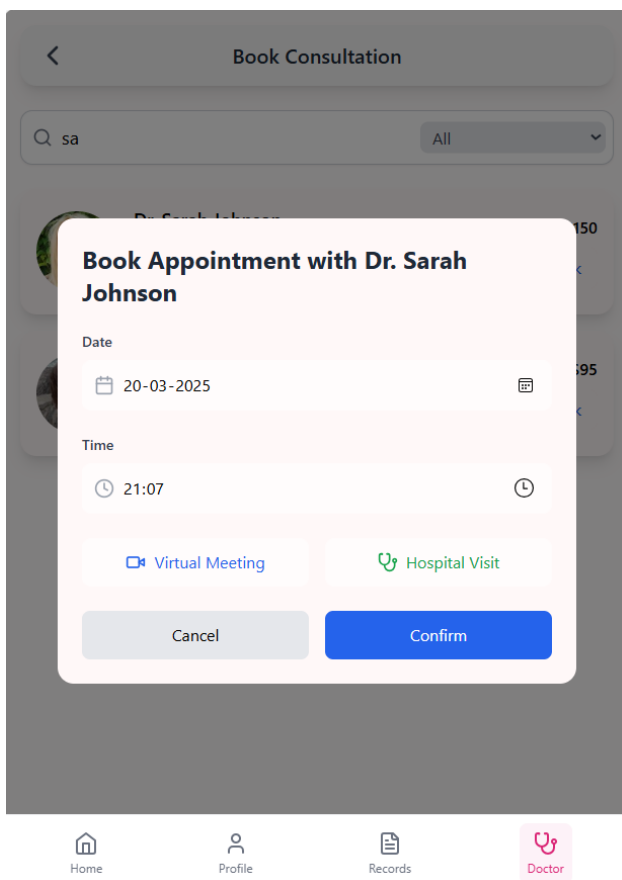


Fig8. Choice to select time, date and mode view

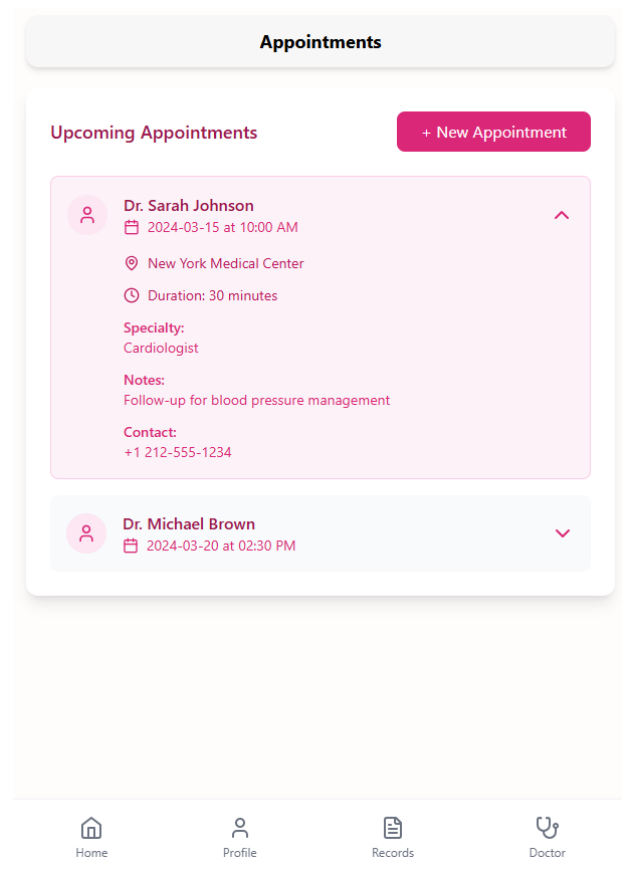


Fig9. Check appointments with a detailed view

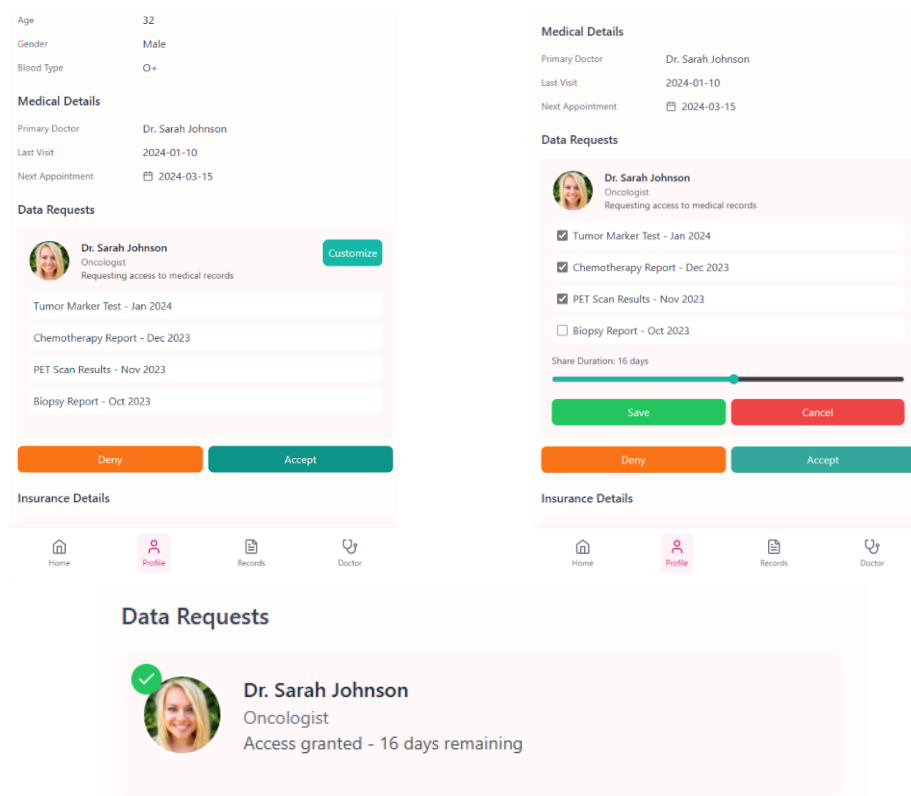


Fig10. Choice of selecting how much data the patient wants to share with the doctor and for how long

- **Featured offered to doctors:**
 - Detailed history of patient before appointment
 - CT scans, MRI, Blood report analysis
 - Tracking appointments
 - Request past health records from patients

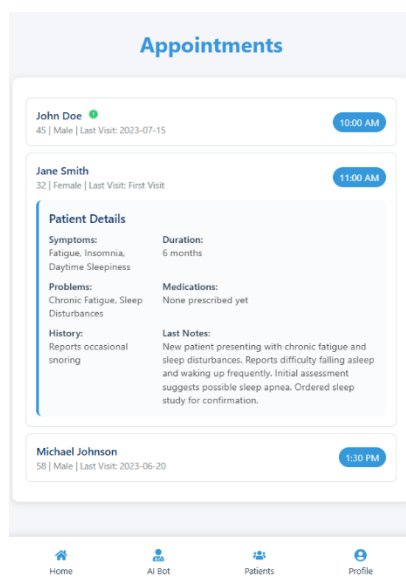


Fig.11 Appointment schedule with Detailed notes for reference

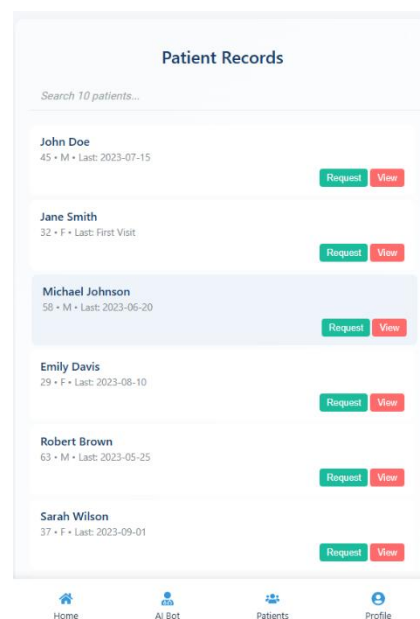


Fig12. List of patients enrolled with the doctor

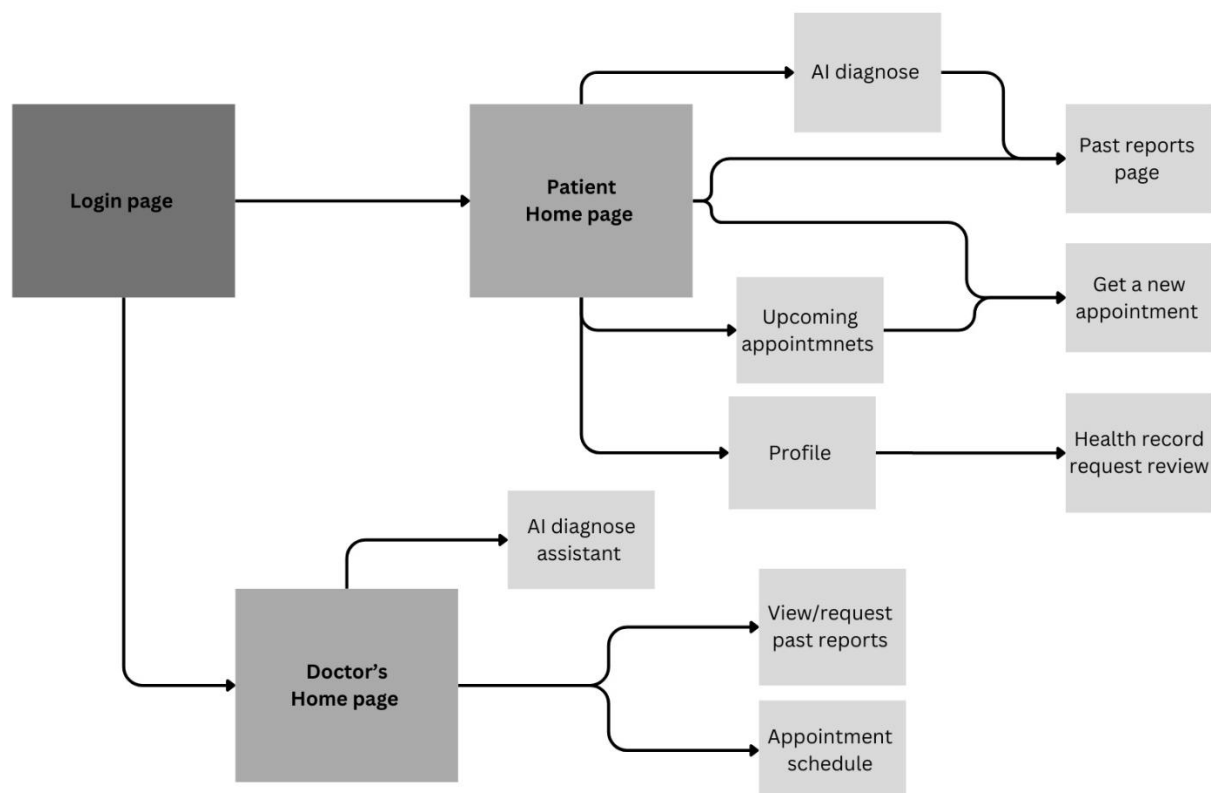
Fig.13 Doctor can request user to share health Reports for a particular time period

Fig14. Health records of user once approved

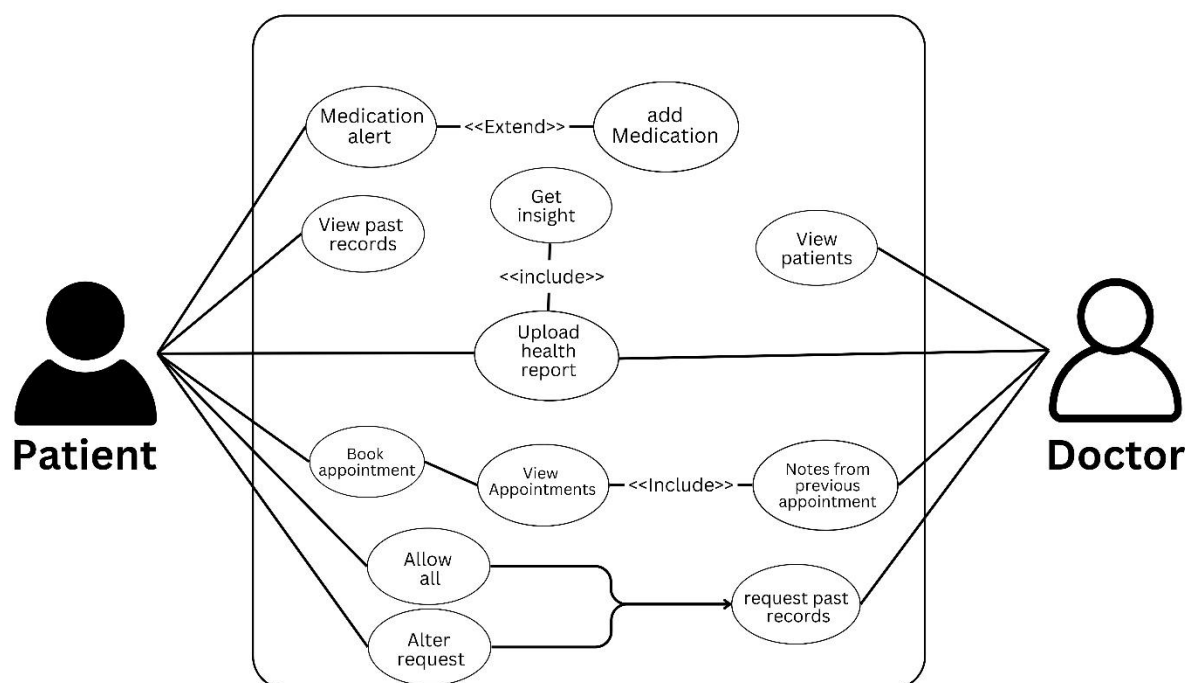
Fig.15 AI bot assistant for doctor to review Reports on their behalf

Fig16. AI generated short summary from the findings of the report

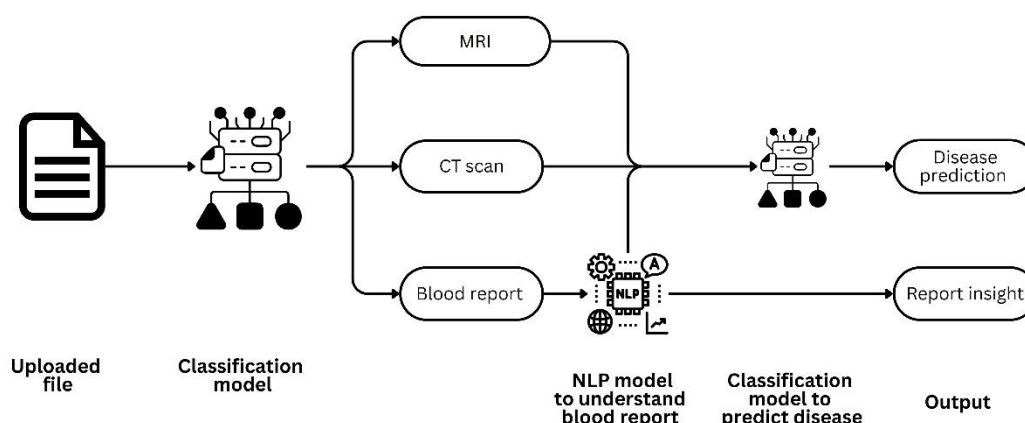
- Page Navigation flow:



- User activity diagram:

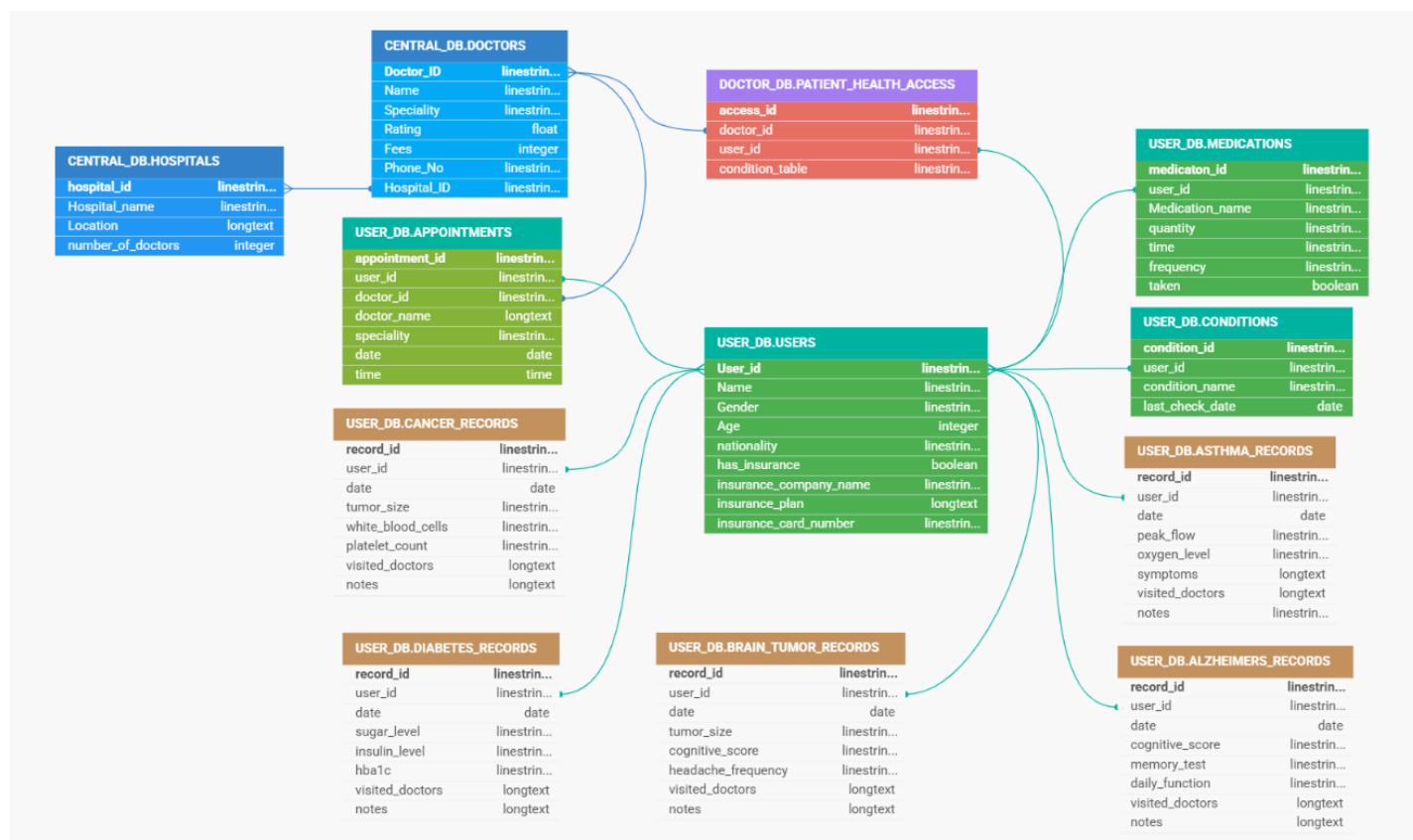


- Structure of AI- classification model

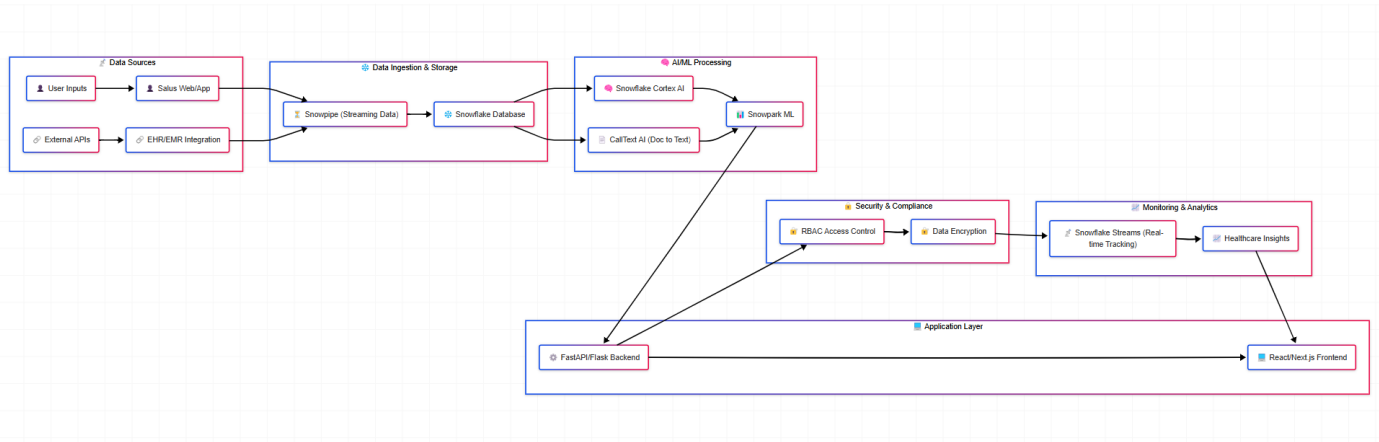


When a report is uploaded it is classified into, CT scan, MRI & blood test report (classes will be increased as further modes of file format are added). If the file is CT scan or MRI it will directly be sent to do disease prediction which can be made from MRI / CT scans. However, if the uploaded file is predicted to be blood report NLP model will be used to extract the parameters from it and based on them, if any disease (diabetes / lung cancer etc.) can be predicted, then it would be processed else a simple insight of the blood report will be generated as shown in *fig.3*

- ER diagram



• Solution Architecture



• Diseases predicted and models used

S.no.	Disease name	Dataset link	Dataset type
1.	Diabetes	https://www.kaggle.com/datasets/mathchi/diabetes-dataset	Text based
2.	Alzheimer	https://www.kaggle.com/datasets/rabieelkharoua/alzheimers-disease-dataset	Text based
3.	Heart-attack prediction	https://www.kaggle.com/datasets/iamsouravbanerjee/heart-attack-prediction-dataset	Text based
4.	Lung cancer	-	Text based
5.	Skin cancer	-	Image based
6.	Skin diseases I. tinea ringworm II. psoriasis III. melanocytic nevi IV. basal cell carcinoma V. dermatitis , melanoma VI. warts VII. eczema VIII. benign keratosis	-	Image based

• Changes Required (8/03/2025)

- Connection to database
- Checking for availability in doctor's schedule
- Inserting login credential page
- Securing database
- Add medicines directly from doctor's interface to user
- Implement add medication option
- Send notification at time of medicine

- **Github Repository Link:** <https://github.com/nischal-smarttech/Salus>
 - **Code directory structure**

```

Salus /
|--- AI_models /
|   |---- Untrained/
|       |--- Script to train models
|       |--- Dataset for models to train
|   |---- AI models
|--- Backend/
|   |--- SQL script to create database
|--- Frontend /
|   |--- Patient_Page/
|   |--- Doctor_Page/
          
```
 - The AI_model folder contains the ai models that will be used in the project, it further has a folder by name “Untrained”, it contains the snowflake script and the dataset for the models which can be trained by uploading the dataset and the script on snowflake
 - The folder “Backend” contains the SQL script for creating and storing few sample values to the database
 - The folder “frontend” has two further folders by the name of “patient_page” and “Doctor_page” which contains the components and code for both the patient and the doctor’s page respectively